

LOCALIZED BOST–CONNES FACTORS ARE AMENABLE: THE FAILURE OF W^* -SUPERRIGIDITY, AND WHAT SURVIVES AT THE C^* -LEVEL

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ABSTRACT. We settle the rigidity question for the type III factors of the localized Bost–Connes systems, and the answer is negative in the strongest available sense.

The groupoid $\mathcal{G}_S = I_S \ltimes \widetilde{\Omega}_S$ has abelian, hence amenable, acting group $I_S = \bigoplus_{p \in S} \mathbb{Z}$, so $M_{K,S,\beta}$ is an injective factor. By the classification of injective factors (Connes, Krieger, Haagerup) its isomorphism class is determined by its flow of weights alone — by nothing beyond the single number λ in type III_λ , by nothing at all in type III_1 . The group $\Sigma_\beta = \text{pr}_{\mathbb{R}}(\Xi_\beta^{\text{ext}})$ of [9] bounds that flow, lying in its point spectrum, and determines it whenever the spectrum is pure point; whatever arithmetic the factor retains is therefore in the flow of weights, and in the cases decided in [9] there is none.

Consequently *class field W^* -superrigidity is false, maximally*. For $S = P_K$ and $0 < \beta \leq 1$ every number field gives type III_1 [9, Thm. 3.2], hence

$$M_{\mathbb{Q},P_{\mathbb{Q}},1} \cong M_{\mathbb{Q},P_{\mathbb{Q}},1/2} \cong M_{K,P_K,\beta} \cong R_\infty$$

for every number field K and every $\beta \in (0, 1]$: the factor remembers neither K , nor S , nor β . Popa’s deformation/rigidity machinery cannot be applied, and not for technical reasons: s -malleable deformations produce rigidity through a spectral gap, which requires the acting group to be non-amenable, and I_S is abelian. Cartan uniqueness holds, but as a consequence of Connes–Feldman–Weiss rather than of Popa, and only up to conjugacy by an automorphism — not up to unitary conjugacy — while orbit equivalence rigidity fails outright, all amenable relations of a given type being orbit equivalent by Krieger.

What survives is at the C^* -level and in inclusions. The obstruction group Ξ_β is an invariant of the C^* -dynamical system $(\mathcal{A}_{K,S}, \sigma)$ through its KMS_β simplex, and is destroyed by taking the weak closure in a single extremal state; this is the operator-algebraic form of the measure-class principle of Paper III of the series. And the symmetry-breaking inclusion $M_{\text{sym}}^G \subset M_{\text{sym}}$ of Paper II has index $|\Xi_\beta|$, so the *pair* recovers the order of the obstruction group even though neither algebra alone recovers anything.

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Sources, code and data for the entire series: <https://github.com/Ruqing1963/localized-bost-connes>; this paper is in [papers/IX-rigidity/](#).

1. AMENABILITY, AND WHAT IT COSTS

Throughout, K is a number field, S an infinite set of finite primes, $\beta > 0$ with $\zeta_{K,S}(\beta) = \infty$, φ an extremal KMS_β state and

$$M_{K,S,\beta} = \pi_\varphi(\mathcal{A}_{K,S})'' = W^*(\mathcal{G}_S, \nu), \quad \mathcal{G}_S = I_S \ltimes \tilde{\Omega}_S, \quad I_S = \bigoplus_{\mathfrak{p} \in S} \mathbb{Z}.$$

The series. The foundational paper is [8] and Paper I is [9]; Papers II, III, V, VII and XXVII, cited by numeral, are in preparation, and nothing proved here depends on them.

Lemma 1.1 (Amenability). *$\mathcal{G}_S$ is a second countable étale groupoid with abelian, hence amenable, acting group; therefore $\mathcal{G}_S$ is amenable, $C_r^*(\mathcal{G}_S) = C^*(\mathcal{G}_S)$, and $M_{K,S,\beta}$ is an injective von Neumann algebra. It is a factor of type III by [8, Prop. 6.1].*

Proof. I_S is a countable abelian group, hence amenable, and a transformation groupoid by an amenable group acting on a locally compact space is amenable; amenability of the groupoid gives injectivity of the associated von Neumann algebra, equivalently the orbit relation of I_S on $(\tilde{\Omega}_S, \nu)$ is amenable and its von Neumann algebra [3] is injective [2]. $\square$

Theorem 1.2 (The complete W^* -invariant). *$M_{K,S,\beta}$ is an injective factor of type III, so its isomorphism class is determined by its flow of weights, and by nothing else:*

- if the type is III_1 , then $M \cong R_\infty$, the unique injective factor of type III_1 [4];
- if the type is III_λ , $0 < \lambda < 1$, then $M \cong R_\lambda$, the Powers factor [1];
- if the type is III_0 , then M is the Krieger factor of its flow of weights, which is a complete invariant [6].

The group $\Sigma_\beta = \text{pr}_{\mathbb{R}}(\Xi_\beta^{\text{ext}})$ of [9, Def. 1.1] constrains the flow of weights: Σ_β is contained in the point spectrum $T(M)$ of the flow [9, Rem. 2.3], so $\Sigma_\beta \neq \{0\}$ excludes III_1 , Σ_β dense forces III_0 , and $\Sigma_\beta = c\mathbb{Z}$ forces $\text{III}_{e^{-2\pi k/c}}$ for some $k \geq 1$ or III_0 [9, Prop. 2.2]. If the flow of weights has pure point spectrum, Σ_β determines it and hence the factor; whether it always does is open. In the two cases decided in [9], namely $S = P_K$ with $0 < \beta \leq 1$ (type III_1 , [9, Thm. 3.2]) and the band set of [9, Thm. 4.2] (type $\text{III}_{e^{-1}}$), the factor is R_∞ and $R_{e^{-1}}$ respectively.

Proof. Lemma 1.1 and the cited classification theorems: Connes classified the injective factors of type III_λ , $\lambda \neq 1$, the III_0 case being reduced to Krieger's theorem on the flow of weights, and Haagerup settled III_1 . The constraints are [9, Prop. 2.2 and Rem. 2.3]; the two decided cases are [9, Thm. 3.2, Thm. 4.2]. $\square$

Remark 1.3 (What the factor can still remember). The one place where arithmetic could survive the weak closure is the flow of weights in type III_0 , or the number λ in type III_λ . Both are functions of the essential-value data of the Radon–Nikodym cocycle $N\mathfrak{a}^{-\beta}$, hence of the sequence of norms together with the measure class of π_β ; by [8, Prop. 6.1] the Connes invariant satisfies $S(M) \subseteq \overline{\Gamma_\beta} \cup \{0\}$ with Γ_β the group generated by the $N\mathfrak{p}^{-\beta}$. What the factor remembers is therefore at most an ergodic flow built from the norm data and the measure class; two systems with the same flow of weights are isomorphic whatever their norms, fields and temperatures, so none of these is reconstructible from the factor.

2. W^* -SUPERRIGIDITY FAILS MAXIMALLY

Theorem 2.1 (No reconstruction). *Let K_1, K_2 be any number fields and $\beta_1, \beta_2 \in (0, 1]$. Then*

$$M_{K_1, P_{K_1}, \beta_1} \cong M_{K_2, P_{K_2}, \beta_2} \cong R_\infty.$$

In particular the isomorphism class of the factor determines neither the number field K , nor the prime set S , nor the inverse temperature β , nor the norms $\{N\mathfrak{p}\}$, nor the symmetry group G_S , nor the obstruction group Ξ_β .

Proof. By [9, Thm. 3.2] the type is III_1 for $S = P_K$ and $0 < \beta \leq 1$, for every number field K ; by Theorem 1.2 the factor is then the unique injective III_1 factor. $\square$

Example 2.2 (Explicit collapses).

K	S	β	type	M
$\mathbb{Q}$	$P_{\mathbb{Q}}$	1	III_1	R_{∞}
$\mathbb{Q}$	$P_{\mathbb{Q}}$	1/2	III_1	R_{∞}
$\mathbb{Q}(i)$	P_K	1	III_1	R_{∞}
$\mathbb{Q}(2^{1/3})$	P_K	0.9	III_1	R_{∞}
$\mathbb{Q}$	band set of [9, Thm. 4.2]	1	$\text{III}_{e^{-1}}$	$R_{e^{-1}}$

The first four are isomorphic von Neumann algebras although their arithmetic input differs in every respect. The fifth is distinguishable from them, but only through the single number e^{-1} , and it has the *same* factor as every other system of type $\text{III}_{e^{-1}}$ — for instance the system of any K and S whose norms are all powers of a single λ_0 with $\lambda_0^{-\beta} = e^{-1}$, when it is of type $\text{III}_{e^{-1}}$ rather than III_0 [8, Prop. 6.1].

Corollary 2.3 (The point spectrum is not an invariant). *The set $\{\text{Np}^{-\beta} : \mathbf{p} \in S\}$ is not a W^* -invariant. Only the closed subgroup $\overline{\Gamma_{\beta}}$ enters Connes' invariant $S(M)$, by [8, Prop. 6.1], and $S(M)$ determines only the type. In particular no reconstruction of the prime norms, and hence no norm-preserving bijection $S_1 \cong S_2$, can be extracted from an isomorphism of factors.*

Proof. The four systems of Example 2.2 with $S = P_K$ have pairwise different norm sets and the same factor. In general $S(M) = \bigcap_{\psi} \text{Sp}(\Delta_{\psi})$ is a closed subset of $[0, \infty)$ determined by the type, and $S(M) \subseteq \overline{\Gamma_{\beta}} \cup \{0\}$ [8, Prop. 6.1]; norm sets generating the same closed subgroup $\overline{\Gamma_{\beta}}$ are indistinguishable by $S(M)$, and by Theorem 1.2 the factor carries only the flow of weights. $\square$

3. WHY POPA'S MACHINERY DOES NOT APPLY

Proposition 3.1 (No spectral gap). *Let $(\alpha_t)_{t \in \mathbb{R}}$ be the s -malleable Bernoulli deformation associated with the product measure $\pi_{\beta} = \bigotimes_{\mathbf{p}} \mu_{t_{\mathbf{p}}}$ on $V = \mathbb{Z}_{\geq 0}^S$. The deformation exists, but the conclusion of Popa's spectral gap argument is unavailable: that argument derives rigidity from the failure of the acting group to have almost invariant vectors, and I_S is abelian, hence amenable, hence has almost invariant vectors in every representation. No subgroup of I_S has property (T) or a spectral gap.*

Proof. An abelian group is amenable and the regular representation of an infinite amenable group has almost invariant vectors; Popa's deformation/rigidity arguments [7] require the opposite, a rigid subalgebra or a spectral gap for the deformation. $\square$

Remark 3.2 (This is not a gap in the exposition). Deformation/rigidity theory exists to produce rigidity where amenability fails. Applied to an amenable algebra it can only recover what Connes' classification already gives, which by Theorem 1.2 is the type. The absence of a Popa-type theorem here is therefore structural, and the correct reading of Theorem 2.1 is not that the rigidity proof is missing but that the rigidity statement is false.

4. CARTAN SUBALGEBRAS AND ORBIT EQUIVALENCE

Theorem 4.1 (Cartan uniqueness, correctly attributed). *$A = L^{\infty}(Y_S, \nu) \subset M_{K,S,\beta}$ is a Cartan subalgebra, and any two Cartan subalgebras of $M_{K,S,\beta}$ are conjugate by an automorphism of M . This is a consequence of Connes–Feldman–Weiss, not of Popa intertwining, and it does not give unitary conjugacy: since $\text{Out}(M)$ is nontrivial for the injective factors, conjugacy by an automorphism is strictly weaker.*

Proof. A is maximal abelian with normalizer generating M , by the groupoid picture, the groupoid being principal on a conull set [8, Prop. 2.8]. A Cartan subalgebra $B \subset M$ gives by Feldman–Moore [3] an ergodic non-singular relation $\mathcal{R}_B$ and a 2-cocycle with $(M, B) \cong (W^*(\mathcal{R}_B, \sigma_B), L^\infty)$; $\mathcal{R}_B$ is amenable because M is injective, hence hyperfinite by Connes–Feldman–Weiss [2], and for hyperfinite relations the 2-cocycle is a coboundary [3]. Two Cartan subalgebras B_1, B_2 thus give two ergodic hyperfinite relations whose associated factors coincide, hence with the same flow of weights; by Krieger’s theorem [6] they are orbit equivalent, so the Cartan pairs (M, B_1) and (M, B_2) are isomorphic, i.e. conjugate by an automorphism of M . Unique Cartan decomposition up to unitary conjugacy is a rigidity phenomenon of non-amenable factors [5], and does not hold here. $\square$

Corollary 4.2 (Orbit equivalence rigidity fails). *An isomorphism $M_{K_1, S_1, \beta_1} \cong M_{K_2, S_2, \beta_2}$ does induce an isomorphism of the underlying orbit equivalence relations, by Theorem 4.1 and Feldman–Moore. But this carries no information: by Krieger’s theorem all amenable ergodic non-singular relations of a given type are already orbit equivalent, so the induced isomorphism exists whether or not the arithmetic inputs are related.*

Remark 4.3 (The Radon–Nikodym cocycle). The cocycle $D(v + a, g; v, g) = \text{Na}^{-\beta}$ is preserved modulo coboundaries by any orbit equivalence, and its essential range is the ratio set. But the ratio set is $S(M) \cap (0, \infty)$, which by Theorem 1.2 is the type. The cocycle therefore transmits exactly the information already counted, and no more; in particular it does not transmit the point spectrum, by Corollary 2.3.

5. WHAT SURVIVES

Theorem 5.1 (Ξ_β is a C^* -invariant, not a W^* -invariant). *The obstruction group $\Xi_\beta(K, S)$ is determined by the C^* -dynamical system $(\mathcal{A}_{K, S}, \sigma)$, through its KMS $_\beta$ simplex $\text{Prob}(G_S/\Xi_\beta^\perp)$ [8, Thm. 3.6]. It is not determined by $M_{K, S, \beta}$, by Theorem 2.1. The passage from the C^* -algebra with its flow to the weak closure in a single extremal state destroys it.*

Proof. The first statement is the classification theorem; the second is Theorem 2.1, since two systems with $\Xi_{\beta_1} \neq \Xi_{\beta_2}$ can have isomorphic factors. $\square$

Remark 5.2 (Consistency with the measure-class principle). Theorem 5.1 is the operator-algebraic form of the principle of Paper III of the series, where Ξ_β is identified as an invariant of the measure class of π_β , all the β -dependence sitting there. Taking the weak closure in a single extremal state retains the measure class only through its Krieger invariant, that is through the type. The present theorem is therefore not a new phenomenon but the same one, seen at the level of von Neumann algebras rather than of cohomology.

Theorem 5.3 (The inclusion remembers the order). *Let Ξ_β be finite, let $M_{\text{sym}} = \pi_{\varphi_{\text{sym}}}(\mathcal{A}_{K, S})'' = \bigoplus_\varepsilon M_\varepsilon$ be the algebra of the symmetric state, the sum over the $|\Xi_\beta|$ extremal KMS $_\beta$ states, and let $M_{\text{sym}}^G \subset M_{\text{sym}}$ be the fixed-point algebra of the action of $G = G_S/\Xi_\beta^\perp$, which permutes the summands simply transitively and identifies M_{sym}^G with any one M_ε embedded diagonally. Then*

$$[M_{\text{sym}} : M_{\text{sym}}^G]_0 = |\Xi_\beta|,$$

the minimal index, so the isomorphism class of the pair determines $|\Xi_\beta|$, even though by Theorem 2.1 neither M_ε nor M_{sym} need determine anything. Equivalently, $\log[M_{\text{sym}} : M_{\text{sym}}^G]_0 = \log |\Xi_\beta|$ equals the relative entropy of an extremal phase against the symmetric one.

Proof. The action of G_S on the KMS $_\beta$ simplex factors through G and permutes the extreme points simply transitively [8, Thm. 3.6(3)], hence the summands of M_{sym} , whose GNS representations are pairwise disjoint; so M_{sym}^G is the diagonal copy $D \otimes 1 \subseteq D \otimes \ell^\infty(G)$ with $D = M_\varepsilon$, whose minimal

index is $|G|$, realized by the averaging expectation. The relative entropy of one extremal state against the uniform mixture of $|G|$ mutually singular states is $\log |G|$. These are the computations of Paper II of the series (its Theorem 5.8 and Corollary 5.11), reproduced here for completeness. $\square$

Remark 5.4 (Where to look for rigidity). Theorem 5.3 indicates the level at which rigidity questions are not vacuous: not the single factor, which is R_∞ , R_λ or a Krieger factor, but structured objects — the C^* -algebra with its flow, the inclusion $M_{\text{sym}}^G \subset M_{\text{sym}}$, the Cartan pair together with the G_S -action. Each of these carries strictly more than the factor, and the first carries Ξ_β in full.

6. AUTOMORPHISMS, AND THE FUNDAMENTAL GROUP

Proposition 6.1 (Both invariants are degenerate). (1) *The fundamental group $\mathcal{F}(M)$ is not defined for type III factors, being an invariant of II_1 factors; the analogous statement, that $M \otimes B(H) \cong M$, holds trivially for every properly infinite factor.*

(2) *$\text{Out}(M_{K,S,\beta})$ is that of the injective factor determined by Theorem 1.2 — of R_∞ or R_λ — and so is the same for all systems of the same type, carrying no arithmetic information.*

Proof. (1) is the definition. (2) is Theorem 1.2: isomorphic algebras have isomorphic outer automorphism groups. $\square$

Remark 6.2 (On the proposed embedding $G_S/\Xi_\beta^\perp \hookrightarrow \text{Out}(M)$). The group G_S acts on the KMS_β simplex, permuting the extremal states simply transitively through $G_S/\Xi_\beta^\perp$ [8, Thm. 3.6(3)]. It therefore acts on M_{sym} , and gives isomorphisms between the different summands M_ε ; but those summands are mutually isomorphic by Theorem 1.2, so the resulting automorphisms of M_ε are not canonically defined and depend on the chosen identifications. We do not claim an embedding $G_S/\Xi_\beta^\perp \hookrightarrow \text{Out}(M_\varepsilon)$ and we do not see how to obtain one; what is canonical is the action on the pair, which is the content of Theorem 5.3.

7. ASSESSMENT

Statement	Status	Reference
$M_{K,S,\beta}$ is injective	true	Lem. 1.1
complete W^* -invariant = flow of weights	true; Σ_β bounds it	Thm. 1.2
class field W^* -superrigidity	false, maximally	Thm. 2.1
reconstruction of β	false	Thm. 2.1
reconstruction of $\{\text{Np}\}$	false	Cor. 2.3
reconstruction of G_S , Ξ_β	false	Thm. 2.1
s -malleable deformation gives rigidity	false: no spectral gap	Prop. 3.1
Cartan unique up to unitary conjugacy	false	Thm. 4.1
Cartan unique up to automorphism	true, by CFW	Thm. 4.1
OE rigidity	false, by Krieger	Cor. 4.2
Ξ_β a C^* -invariant	true	Thm. 5.1
$[M_{\text{sym}} : M_{\text{sym}}^G]_0 = \Xi_\beta $	true	Thm. 5.3
$\mathcal{F}(M)$, $\text{Out}(M)$ carry arithmetic	false	Prop. 6.1

The picture is uniform with the rest of this project. In Paper III of the series the arithmetic was shown to be an invariant of a measure class, invisible to K -theory, to groupoid homology and to dynamical entropy, all of which discard the Radon–Nikodym derivative. Passing to the von Neumann algebra discards it too, retaining only its Krieger invariant; and since the groupoid

is amenable, Connes' classification then leaves nothing at all. The failure recorded here is not a failure of technique. It is the same measure-class principle, applied one level further out.

Three questions. First, is there a non-amenable enlargement of the system to which deformation/rigidity does apply? Paper V of the series shows that a free non-abelian semigroup empties the high-temperature region, and Paper XXVII that non-abelian symmetry can be obtained with the abelian group I_S acting through a non-abelian cocycle — but the groupoid is then still amenable, the factor still injective, and nothing here changes; an Ore semigroup that is non-amenable but retains nested range projections would be the object to look for, and we do not know whether one exists. Second, how much does the Cartan pair together with the G_S -action remember? Theorem 5.3 recovers $|\Xi_\beta|$ from the inclusion; whether the triple (M, A, G_S) recovers Ξ_β itself, or even S , is open. Third, at the C^* -level the classification of [8] is complete, but we have not asked whether $\mathcal{A}_{K,S}$ itself, without its flow, remembers anything; by Paper II its K -theory does not, and the question of C^* -isomorphism is untouched.

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